PHONE-BASED CATARACT SCREENING ON FIELD IMAGES: TECHNICAL REPORT

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Phone-Based Cataract Screening on Field Images: Technical Report

PROJECT REPORT

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Summary

In Indian charity eye hospitals, a few ophthalmologists face very large patient numbers, and the doctor's examination time decides how many people are screened for cataract. Drawing on four years of field work with Vivekananda Mission Charitable Hospital, Midnapur, West Bengal in remote India, this work aims at a tool that works on the images a busy clinic actually produces: phone photographs of undilated eyes, hand-held, with flash, one attempt per eye. Once build, the more valuable application wld be to install in village Kiosk, that can help villager to regularly self-screen, as it is difficult for them to visit hospital regularly.

On 744 clinically labelled eye crops from such photographs, a standard classifier reached AUC 0.69, and three different changes did not move it. Error analysis with the clinician found the reason: under flash, corneal glare and lens opacity look alike in the pupil. Controlling capture with retakes can avoid the problem, but it is rarely practical in a high-volume clinic. The direction taken here is a model that reads through field errors the way a clinician does. This report sets out the computer-vision path for it. The work is in progress.

1. Problem and objective

The bottleneck in cataract screening at a high-volume charity hospital is the doctor's examination time. Over several years of field work with Vivekananda Mission Charitable Hospital, screening delays were observed to send patients home unassessed. If a paramedic could photograph each eye and a model could score it, patients would reach the doctor already ordered by likelihood of cataract and also help in remote check ups in Kiosks.

In such a clinic, images are taken quickly, by hand, under existing light, usually once per eye. Image errors are therefore a normal property of the input, not exceptions:

Table 1. Field conditions and their effect on the image.

Field condition	Effect on the image
Hand-held, one attempt per eye	Blur, variable distance and zoom; no second view
Phone flash, ward lighting	Specular spots and broad corneal reflections over the pupil
Undilated pupil	Small pupil, so less of the lens is visible; flash makes clouding easier to see but adds reflections

Field condition	Effect on the image
Operator-dependent framing	One eye, a full face with both eyes, or off-centre gaze
Sharing via messaging apps	Compression and duplicate copies

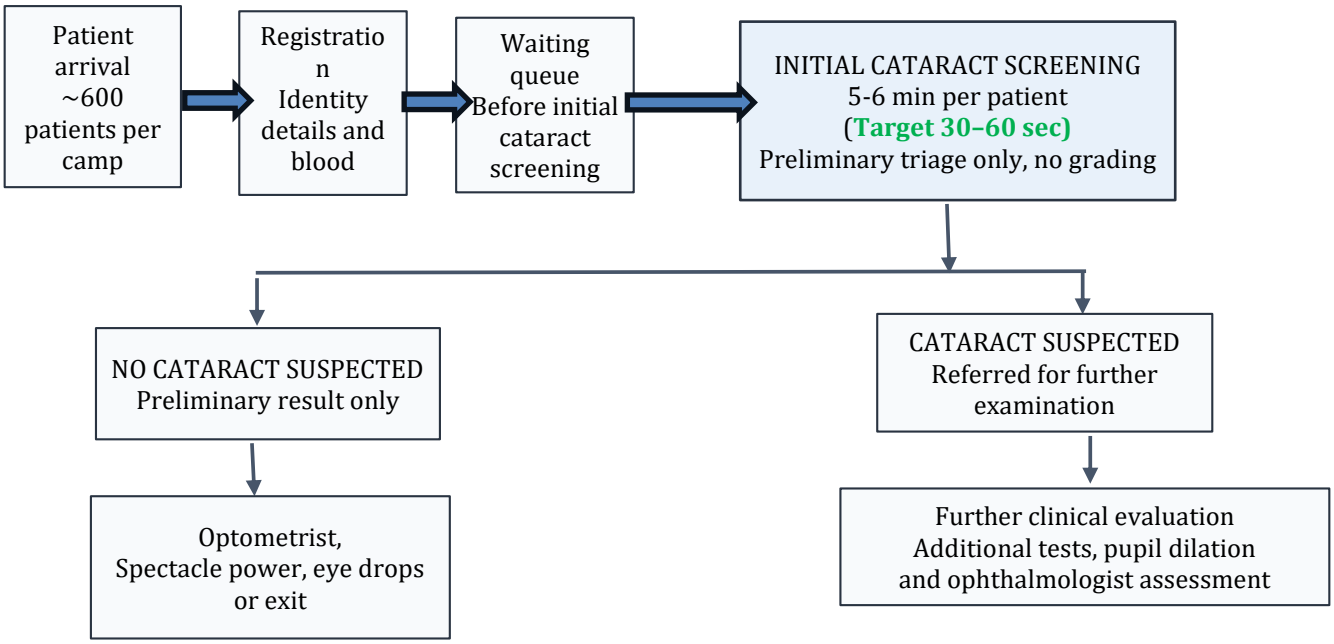
In the collected data, 43% of photographs showed both eyes, and 739 of 744 eye crops contained at least one flash reflection.

Field constraints that shape the design. These come from four years of work with the hospital, observing what is and is not adopted in remote clinics. Paramedics work against a queue and take one photograph per eye; any step that adds time, such as a retake prompt, tends to be skipped. Phones, lighting and operators vary from day to day. A tool is therefore useful only if it accepts the image as taken and adds no step to the workflow.

Objective. A tool that works on these images as they are. This work measures how far a standard approach gets on them, finds what limits it, and identifies the technical path to remove that limit.

2. Clinical background

The patient flow below is from screening camps run by Vivekananda Mission Charitable Hospital, Midnapur, West Bengal. A camp sees about 600 patients between 8:30 a.m. and 2:00 p.m., with a fixed number of technicians. After registration, the initial cataract screening counter decides only whether cataract is suspected, with no grading. At about 5-6 minutes per patient this counter is the bottleneck, using roughly 30 staff-hours per camp; cutting it to 30–60 seconds with a phone-based check would free 25-25 staff-hours and could raise capacity towards 1,000 patients per camp.



Timing interpretation
At 5-6 min each: 600 screenings = 30 cumulative staff hours.
At 30–60 sec each: 5–6 cumulative staff-hours. Goal: more patients per camp.

2.2 Cataract types and grades

A cataract is a clouding of the eye's lens. Doctors distinguish three main types by where the clouding sits:

Table 2. Main cataract types.

Type	Where	How it looks
Nuclear sclerotic (NS)	Centre of the lens; the most common type	Diffuse whitish to yellowish clouding in the middle of the pupil
Cortical	Outer lens, extending inward	Spoke- or wedge-shaped opacities
Posterior subcapsular	Back surface of the lens	A cloudy or patchy area

Nuclear cataract is graded by severity, informally NS 1 to NS 4, with "NS 4++" used for very advanced cases. Discussions with the hospital's doctors, while reviewing this project's photographs, gave the following picture of what a phone photograph can show:

Table 3. Nuclear cataract grades and what a phone photograph shows (doctors' observation).

Grade	Severity	Visible in a phone photograph? (doctors' observation)
NS 1	Mild, early	Rarely; usually needs a slit-lamp examination
NS 2	Noticeable	Sometimes; doctors recognised some of our photographs despite glare
NS 3	Advanced	Generally yes, even with glare
NS 4 / 4++	Very advanced	Readily

Cortical and posterior subcapsular cataracts are harder to recognise in these photographs; only advanced cases may show. On the doctors' advice, **this work focuses first on nuclear cataract**.

Implications for the model -

- **A realistic target is NS 2 and above:** NS 3 and NS 4 are within reach of a photograph; NS 1 largely is not, whatever the model. NS2 is possible most tiems by a doctor's experiences eye.
- **The labels carry no grade:** The dataset is labelled cataract or non-cataract from clinical examination. Cataract eyes include early grades that the photograph may not show, which caps any image-based score. Recording the grade at capture would allow results per grade.
- **Doctors see a usable cue through glare:** In NS 2 photographs with glare, the cataract region looked slightly blurred, matte and off-white or yellowish, while flash reflections were crisp and pure bright white. Colour and sharpness are therefore candidate features alongside the pupil boundary (Sections 6 and 8).

- **Screening is not a surgery decision:** Whether to operate depends on a full examination of the eye, not only on cataract maturity. The tool's role is to order or refer patients, never to replace that examination.

3. Data

The images were collected on site at camps of Vivekananda Mission Charitable Hospital during routine screening, without any capture protocol. The treatment ophthalmologist labelled each photograph as cataract or non-cataract: 817 photographs in total (298 cataract, 519 non-cataract) in three batches.

Photographs showing both eyes were reviewed eye by eye with the ophthalmologist, so each eye has its own label. Both eyes of one photograph always stay in the same data split.

After duplicate removal and quality control, the final set is **744 eye crops (255 cataract, 489 non-cataract)**. Before any training, 20% of photograph groups (148 eyes, 51 cataract) were locked away as a test set. The remaining 596 eyes are used with 5-fold group cross-validation.

[Figure placeholder: panel of six eye crops from the dataset: clear cataract; normal eye with glare; cataract with glare; two-eye photograph; side gaze or blur; a crop removed in quality control.]

4. Method

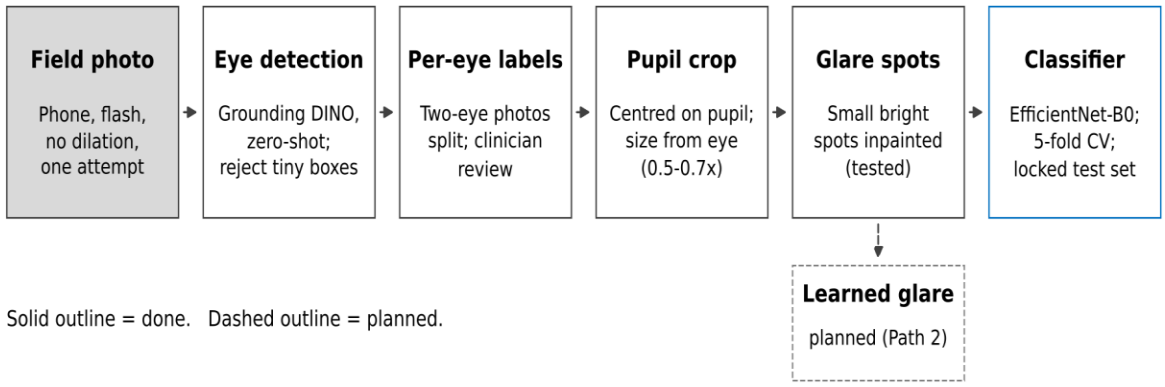


Figure 1. Processing pipeline from field photograph to eye-level cataract score. Solid boxes are done; the dashed box is planned.

Pipeline: The upload was verified byte for byte. Duplicates were removed with perceptual hashing at photograph level (pHash, distance ≤ 5) and again at crop level. Eyes were found with a zero-shot detector (Grounding DINO), giving 349 single-eye and 289 two-eye

photographs. After clinical review, one crop per usable eye was cut (783), then checked visually on contact sheets, and 39 non-eye or glare-only crops were removed.

Cropping needed three versions, each driven by an observed failure:

Table 4. Cropping versions and the failure that drove each change.

Version	Method	Observed failure
1	Square crop at 1.4× the detected pupil box	Detector sometimes boxed a flash reflection as the pupil: crops of glare only
2	Centred on the pupil, sized from the eye box (0.5–0.7× eye width)	Remaining glare-only crops where the "eye" box was itself a reflection
3	As 2, rejecting eye boxes under 8% of the image side	Zoom still varies with gaze and distance; accepted

Training. EfficientNet-B0 (ImageNet-pretrained), one output, BCE loss, AdamW with a cosine schedule. Augmentation: random crop, flip, ±15° rotation, brightness and contrast ±20%; no hue change, because lens colour is a clinical cue. The decision threshold was set for 95% sensitivity. Four experiments each changed one factor:

Table 5. Experiments: each changes one factor.

Experiment	Change	Question tested
Baseline	Eye crops, 384 px, learning rate 2e-4, 20 epochs	Is there a learnable signal?
Whole photo	Full photographs instead of eye crops	Does cropping lose information?
Slow learning	Lower learning rate (5e-5), 12 epochs	Is training too aggressive?
Glare-spot removal	Small glare spots inpainted, 512 px, dropout 0.3 and weight decay 1e-2	Do glare spots and overfitting limit it?

5. Results

Figure 2. Fold-level and cross-validated AUC for each experiment. (Whole photo was stopped after three folds and is not shown.)

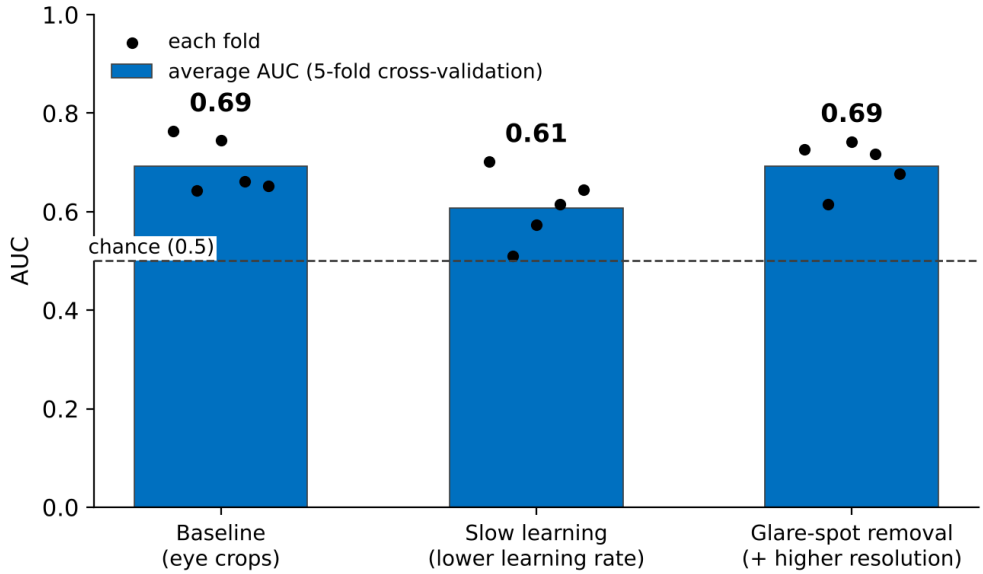


Table 6. Results on 596 development eyes, 5-fold cross-validation.

Experiment	CV AUC	Sensitivity / specificity at threshold
Baseline	**0.692** (95% CI 0.648–0.734)	0.96 / 0.18
Whole photo	0.57–0.77 per fold (stopped after 3)	—
Slow learning	0.608	—
Glare-spot removal	**0.693**	0.96 / 0.14

AUC: probability that a random cataract eye scores above a random normal eye (0.5 = chance). Sensitivity: share of cataract eyes flagged; specificity: share of normal eyes cleared.

In the baseline, training loss fell from 2.19 to 0.05 over 20 epochs while validation AUC stayed at 0.70–0.76 and peaked in the first epoch: the model memorises training images without improving on new eyes. Whole photographs were not better than crops, a lower learning rate was worse, and removing small glare spots changed nothing. **Performance plateaus at about 0.69.** The locked test set was not used, so that it remains an unbiased check for a later model.

6. Performance stops at 0.69

Two physically different things produce similar whitish regions in the pupil: **lens opacity** (light scattered inside the lens, behind the iris plane) and **corneal reflection** (light

reflected at the corneal surface, in front of it). The ophthalmologist separates them reliably in the same images. Error analysis with the clinician gave these cues:

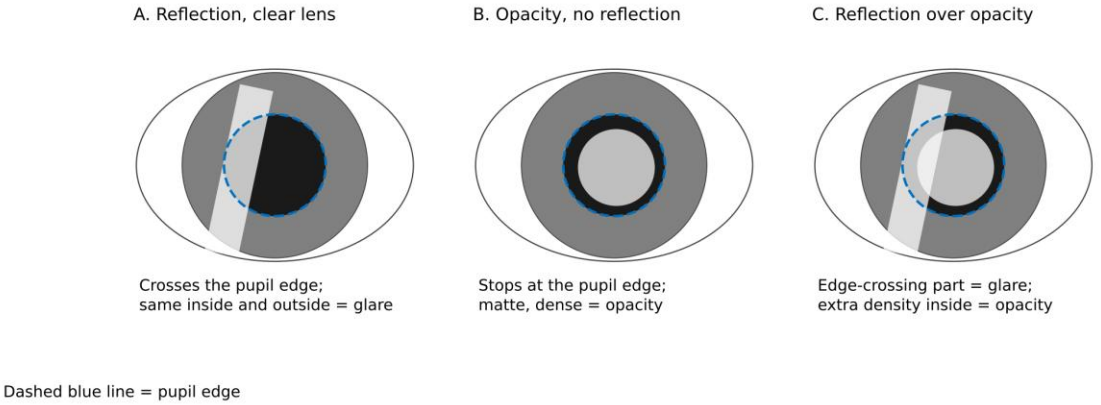


Figure 3. The spatial cue: reflection crosses the pupil edge; opacity is bounded by it

Table 7. Cues that separate lens opacity from corneal reflection.

Cue	Lens opacity	Corneal reflection
Extent	Confined to the pupil	Can cross the pupil edge onto the iris
Inside vs outside the pupil	Denser inside	Similar inside and outside
Character	Density: matte, milky	Brightness: specular, crisp
Colour and edge	Off-white to yellowish; slightly blurred	Pure bright white; sharp

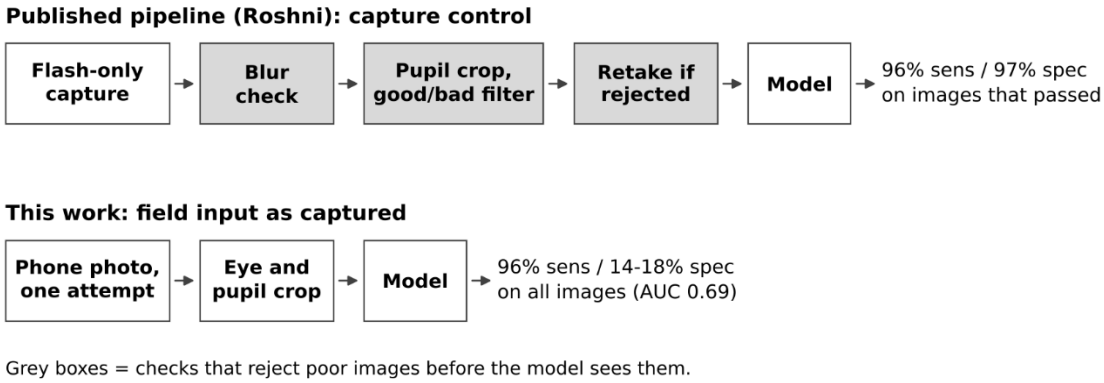
Three refinements matter for any algorithm. **Brightness is not density**: an opacity can be dull grey-white, so brightness thresholds find glare, not cataract. **Reflection over a clear lens also looks cloudy** where it covers the pupil. **Reflection can sit on top of an opacity**, so naive removal can remove the cataract. The most reliable single cue is spatial: whiteness continuing past the pupil edge is reflection. This explains the glare-spot removal result: spot removal acted on brightness and size, not on the pupil boundary. The limit is therefore not the classifier but what it is able to see in the image.

7. Two ways to handle field image quality

With performance stopping at 0.69, we checked how published systems perform. The closest is the Roshni study from Uttar Pradesh: a smartphone app for frontline health workers, developed on about 13,600 flash images of eyes and field-tested on 604 eyes. It reports 96% sensitivity and 97% specificity in development, and 86.6% and 93.3% in the field. Its main difference from this work is where image quality is handled: at capture, before the model, or inside the model.

Capture control. The Roshni study takes the first route. Each image passes automatic checks: a blur check (variance of the Laplacian, an edge-sharpness measure, rejecting values below 150), a good/bad iris classifier, and a good/bad pupil classifier trained to reject pupils covered by flash. A rejected image is not repaired; the app asks the health worker to retake it. The model was developed only on images that passed these checks and two-doctor agreement (2,779 of 4,000).

Technically, this success depends on retakes. The checks make each field image resemble the clean images the model was trained on. How often retakes were needed is not reported. Field experience in heavy-volume hospitals says retakes are rarely practical: each extra capture



slows the queue, and paramedics avoid them (Section 1).

Figure 4. Capture control versus no control: the published pipeline gates and retakes images before the model; this work scores every image as captured.

A robust model

The other route is a model that handles field images as they are, reading through glare and blur as a clinician does. This is mainly a computer-vision problem.

8. Computer vision processing

This work is in progress, and its direction is decided. A method that buys accuracy with retakes slows the queue, and in a high-volume hospital the queue is the problem being solved. The same issue appears in the planned village kiosk: retakes are possible there, but a villager who cannot hold the eye at the right angle will be asked to retake again and again, and may give up. Neither setting can depend on a perfect image.

The model must therefore learn the error types of field images and become robust to them, instead of having them filtered out. A robust model serves both the hospital and the kiosk. The methods below, grouped by the clinical cue from Section 6, are the route to it:

A. Find the pupil edge precisely (needed for the spatial cue)

1. *Circular edge detectors:* the Hough circle transform and Daugman's integro-differential operator from iris recognition locate the pupil boundary in noisy eye images. Pupil outlines already exist for all 744 crops (728 found directly, 16 via the iris) and can be refined this way.

B. Separate reflection from opacity (brightness versus density)

2. *Specular–diffuse separation:* the dichromatic reflection model splits an image into a specular part (mirror-like glare, the colour of the light) and a diffuse part (the material itself). This matches the doctors' cue: glare is the white of the flash, while nuclear opacity has its own off-white to yellowish colour. Single-image methods (Tan and Ikeuchi) are a candidate; untested on eyes.
3. *Colour and sharpness features:* measure yellowness (for example the b* channel of the Lab colour space) and edge sharpness inside the pupil, to separate matte, yellowish opacity from crisp white glare. These can be given to the model as extra inputs or used to check what it learns.
4. *Learned reflection separation:* networks that split a photograph into a transmission layer and a reflection layer (e.g. Zhang et al., 2018). Built for glass; untested on eyes.
5. *Boundary-aware glare rule (this work):* remove whiteness only outside the pupil, leaving the interior untouched. Implemented and tested on synthetic eyes; not yet trained on.

C. Train the model the error types

6. *Glare as a second label (multi-task):* predict cataract and "reflection crosses the pupil edge" together, with the second label generated automatically from pupil outlines (related: Che et al., 2023).
7. *Synthetic reflections in training:* add realistic, edge-crossing reflections to both classes, creating the "same eye, different glare" pairs the data lacks (related: Daher et al., 2023; Byrne et al., 2023).
8. *Mixed training* on processed and original crops together, which the Mo-Cat study found better than either alone; *zoom augmentation* over the observed capture distances.

D. Check what the model looks at

9. *Grad-CAM attention maps*, to see whether the model attends to the pupil interior, the reflection or the iris; Mo-Cat found a full-eye model attending to skin.

Each method will be compared with the current model on the same locked test set, and, once grades are recorded, per NS grade. More field data (one eye per photo, flash on and off) would supply the missing glare/no-glare pairs.

9. Conclusion and Limitation

Use now - At AUC 0.69 the model suits only queue ordering: every patient is still examined and only the order changes. Clinicians indicated about 70% accuracy would already help here; it cannot be used to turn anyone away. The doctors' longer-term aim is a local-language village kiosk for self-screening, saving repeated hospital trips; it differs technically (self-capture without a trained operator, very high sensitivity needed) and requires its own validation data. Even at the current accuracy, such a kiosk could act as an early nudge, prompting villagers whose photographs score high to visit the hospital, where some false positives are acceptable. If the model becomes robust, the kiosk is its major application, for villagers who cannot spare a day for the hospital trip.

On field images as captured, EfficientNet-B0 reaches AUC 0.69, stable across changes to input, training and preprocessing. The limit is identified as, under flash, glare and lens opacity look alike, and the model has not learned the boundary and density cues a clinician uses. Because retakes are not practical in a high-volume clinic, the solution has to be a model robust to field images. This work is in progress and continues in this direction, starting with the glare label, pupil-edge cues and synthetic reflections.

Limitations - 255 cataract eyes is small for a subtle visual distinction. All images come from one hospital and one period. In two-eye photographs the eye not named as cataract is labelled normal, and single-eye photographs keep their photograph label without per-eye review. Labels carry no cataract grade, and early grades (NS 1) are rarely visible in photographs at all. One batch contains only cataract images shared through a messaging app and is monitored for bias. Results are cross validated; the locked test set is unused by design.

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